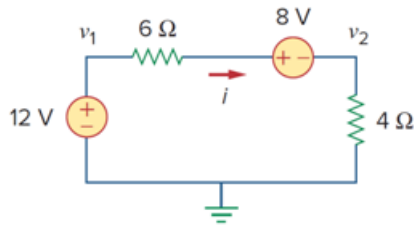


Problem

In the circuit of Fig. 3.47, the voltage v_2 is:

- (a) -8 V
- (b) -1.6 V
- (c) 1.6 V
- (d) 8 V

Figure 3.47



Step-by-step solution

Step 1 of 5

Refer to figure 10.46 given in the textbook.

Calculate the impedance of inductor.

$$Z_L = j\omega L$$

Substitute 1 for ω , and 1 for L .

$$\begin{aligned} Z_L &= j(1)(1) \\ &= j1\ \Omega \end{aligned}$$

Step 2 of 5

Calculate the impedance of capacitor.

$$Z_C = \frac{1}{j\omega C}$$

Substitute 1 for ω , and 1 for C .

$$\begin{aligned} Z_C &= \frac{1}{j(1)(1)} \\ &= \frac{1}{j} \\ &= -j1\ \Omega \end{aligned}$$

Step 3 of 5

Consider the voltage source.

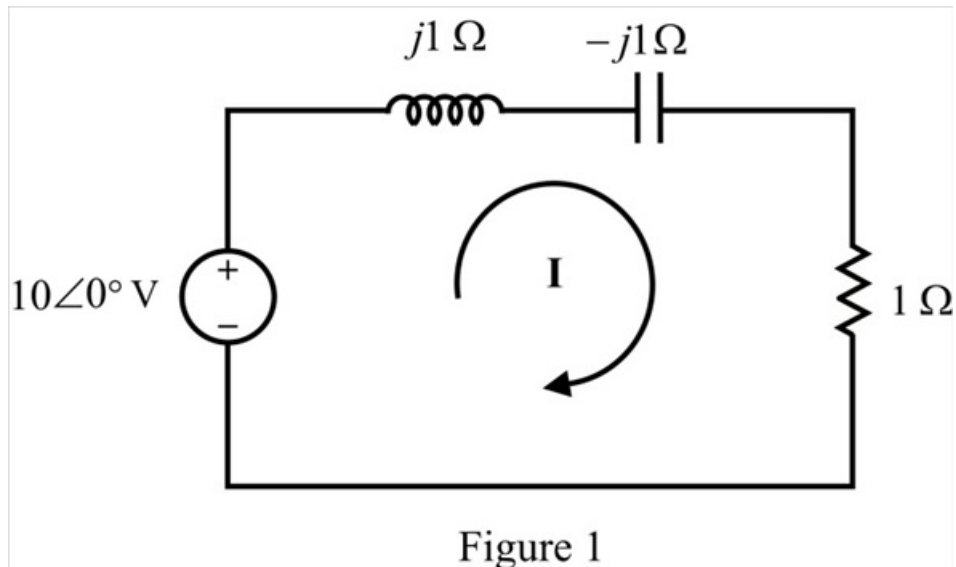
$$v(t) = 10 \cos t\text{ V}$$

Convert $v(t)$ into phasor form.

$$\mathbf{V} = 10 \angle 0^\circ\text{ V}$$

Step 4 of 5

Redraw the circuit with the impedances of inductor and capacitor as shown in figure 1.



Step 5 of 5

Here, current $\mathbf{I}$ is the phasor form of current $i(t)$.

Apply Kirchhoff's voltage law to the circuit shown in figure 1.

$$10\angle 0^\circ - (j1)\mathbf{I} + (j1)\mathbf{I} - \mathbf{I} = 0$$

$$\mathbf{I} = 10\angle 0^\circ \text{ A}$$

Convert current $\mathbf{I}$ into time domain.

$$i(t) = 10\cos t \text{ A}$$

Therefore, the value of current $i(t)$ is $10\cos t \text{ A}$.

Hence, the correct option is **(a)**.